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A Real-Time Haptic Simulation Prototype for Liver Lesion Localization Training: System Design, Technical Feasibility Validation, and Formative Expert Evaluation

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ABSTRACT Haptic feedback has potential value in medical training, but the technical feasibility of integrating continuum physics computation into a real-time haptic loop remains insufficiently explored. This study is framed as a system-integration and technical-feasibility study: it develops a task-oriented haptic training prototype to evaluate the system connectivity of a physics engine in an interactive palpation task. Using liver lesion localization as the application scenario, we adopt a group-based corotational finite element method as the physical model, combine it with bimanual tracking and a fingertip-worn interface, and establish a closed loop between large-deformation computation and force response. Hidden lesions are simulated through local modulus enhancement, and the extracted contact-normal reaction force is mapped to haptic output. All runtime tests were conducted on consumer-grade hardware, and a formative evaluation with one surgical expert was performed to identify engineering boundaries; this single-participant walkthrough provides diagnostic, exploratory feedback and does not by itself support claims of clinical-grade training efficacy, educational impact, or learning outcomes. The results show that the prototype supports system-level operation driven by physics computation and demonstrate the technical feasibility of this haptic integration architecture and device deployment. Based on these findings, the discussion highlights the engineering challenges that remain before the current physical model could be translated into a clinical-grade training system, including simplified anatomical constraints and the absence of nonlinear material properties, thereby providing design implications and engineering references for future medical palpation systems.

INDEX TERMS Liver palpation, real-time simulation, corotational finite element method, haptic interaction, medical training.

I. INTRODUCTION

Haptic feedback and soft-tissue simulation have long been regarded as components of medical training systems, but their effectiveness does not depend only on whether haptics is present. It is closely related to task type, evaluation method, and feedback design. Existing reviews show that the value of haptic feedback is often more evident in complex tasks, while training outcomes are still influenced by device degrees of freedom, feedback realism, and experimental design [1]–[3]. For liver lesion localization, operators must compare local stiffness differences under limited visual information. This makes the task a natural testbed for asking whether physics computation can be translated into perceivable haptic cues. Prior work has already shown that liver palpation diagnosis can itself serve as a standalone simulation target, and that liver-related interactive systems are not limited to resection or cutting tasks [4], [5].

From the computational perspective, soft tissues such as the liver exhibit pronounced large deformation and near-incompressible behavior during pressing and traction. Real-time simulation therefore faces a dual constraint of accuracy and efficiency. Continuum finite element methods (FEM) offer strong physical interpretability and are well suited to representing local stiffness changes and material-parameter differences, but traditional global solvers are expensive. To address this issue, prior research has explored corotational FEM, cutting FEM, and grouped solving strategies to preserve physical credibility under real-time constraints [6], [7]. At the same time, Projective Dynamics, position-based dynamics (PBD), and extended position-based dynamics (XPBD) offer advantages in interactive speed, numerical stability, and implementation convenience [8]–[12]. However, when the goal is to represent lesion stiffness rather than merely visually plausible motion, material interpretability and mechanical consistency remain central.

From the interaction perspective, high-degree-of-freedom (DOF) force-feedback devices can provide richer kinesthetic and contact information, but they usually come with engineering costs such as structural complexity, high price, and restricted workspace [13]. Review studies likewise indicate that medical training systems do not necessarily benefit from blindly pursuing the highest number of degrees of freedom or the highest possible fidelity. What matters more is whether the critical perceptual cues are emphasized for a specific task [1], [2]. Recent work also stresses task-oriented force-profile design rather than reconstructing every haptic dimension by default [14]. This provides theoretical support for the one-degree-of-freedom (1-DOF) interface used here: if lesion search depends primarily on local differences in normal resistance, then low-complexity hardware may still play a meaningful role in preclinical validation.

From the educational perspective, training for practical medical skills such as palpation faces challenges of real-case shortages and ethical risks, and practical teaching therefore needs a reproducible practice environment that offers rich, dynamic feedback. Chernikova *et al.* [15] note that simulation-based learning has positive utility in higher education and that sufficient practice volume is a prerequisite for developing professional skill, while Wisniewski *et al.* [16] report that the information content of feedback directly shapes learning achievement. Motivated by these two needs, this study builds a real-time physical computation loop on consumer-grade hardware, aiming to remove the physical-space and equipment constraints of traditional simulators and to close the loop between FEM-based large-deformation computation and force response, so that fingertip feedback tracks the modulus changes of internal liver lesions.

Overall, existing studies provide foundations in real-time soft-tissue modeling, liver-related training tasks, and medical haptic-system design, but relatively few works integrate a physically meaningful continuum model, interactive large-deformation updating, and a haptic loop into a single task-oriented system that addresses both needs above. Although previous research has individually explored algorithmic frameworks (e.g., GB-cFEM [7]) and lightweight haptic hardware [17], the system-level integration of these modules faces technical challenges, including maintaining real-time stability under large deformations, preserving near-incompressible volume constraints, and ensuring synchronized force-feedback mapping on consumer-grade hardware.

Based on this motivation, this paper proposes and implements a task-oriented haptic training prototype to evaluate the system connectivity and engineering feasibility of integrating continuum physics computation into a lightweight haptic loop. The incremental contribution of this work lies not in a new standalone algorithm or device, but in overcoming the integration challenges above to build a closed-loop system that responds to these educational needs. The prototype uses liver lesion localization as a concrete test scenario, adopts a group-based corotational finite element method (GB-cFEM) as the core physical engine, combines it with Stereo IR-170 bimanual tracking and a haptic interface, simulates hidden lesions via local modulus enhancement, and maps contact-normal reaction force to haptic output. To provide spatial constraints without compromising real-time performance, the system utilizes a simplified abdominal collision model, acting as a static envelope shell generated from the rest-pose geometry to represent simplified surrounding organs. The evaluation in this paper is a formative expert evaluation whose goal is to identify system feasibility and engineering boundaries. The main contributions are as follows:

1) A task-oriented GB-cFEM physical-haptic closed-loop system prototype is proposed and implemented, enabling real-time deployment on compact consumer-grade hardware that supports the hardware-accessibility need described above.
2) A haptic mapping and exploration mechanism for lesion localization is designed. By extracting contact-normal forces from a continuum physics engine and translating them into dynamic force output via a fingertip-worn interface, the system conveys regional hardness differences to assist learners in interpreting internal lesion mechanics.
3) A formative evaluation involving a surgical expert is conducted. Through walkthrough-based assessment, the study evaluates the prototype's ability to transform FEM-based stiffness into recognizable mechanical cues and identifies the engineering boundaries and practical constraints that remain before translation to a clinical-grade trainer.

II. RELATED WORK

Interactive simulation systems for surgical training and palpation-skill acquisition typically require engineering tradeoffs among visual rendering rate, haptic update frequency, and physical-modeling complexity. Reviews in this area note that virtual-reality and haptic-feedback systems for medical training are limited not only by tissue-mechanics modeling and real-time rendering, but also by task definition, evaluation methodology, and the maturity of educational evidence [1]–[3]. For the liver palpation training task studied here, the most relevant research threads concern real-time soft-tissue modeling, liver-related training tasks, and low-dimensional haptic rendering.

A. REAL-TIME SOFT-TISSUE MODELING FOR SURGICAL SIMULATION

In soft-tissue modeling for surgical simulation, continuum FEM is a standard class of methods because it naturally represents material parameters, boundary conditions, and volumetric mesh structure, although it becomes computationally expensive under large deformation, near incompressibility, contact, and cutting. Corotational FEM separates large rotation from strain computation and achieves a favorable balance between efficiency and deformation quality, and has been used in real-time surgical settings such as needle insertion [6]. Most directly relevant to this work is GB-cFEM, which combines grouped solving, domain decomposition, and offline precomputation to support large tetrahedral models, anisotropic materials, and near-incompressible cases on multicore CPUs [7]. This makes it especially suitable as the core of the liver large-deformation simulation used here.

Beyond traditional FEM, Projective Dynamics and its extensions for hyperelastic materials also provide a balance between stability and real-time performance [8], [9], while soft-tissue models based on constraint energies further improve physical consistency in tool-tissue interaction [18]. By comparison, PBD and XPBD are often used as engineering baselines in real-time systems because of their implementation simplicity and numerical robustness [10]–[12]. However, when anisotropy, near-incompressible volume preservation, and organ-level physical interpretability are required, FEM generally remains the more appropriate primary model.

B. LIVER-RELATED SIMULATION SYSTEMS AND TRAINING TASKS

In interactive simulation for the liver and other abdominal organs, previous studies span organ deformation and cutting workflows, as well as palpation-based lesion exploration and training. Prior work has reported a liver-parenchymal transection system with haptic feedback [5], while another study presented a haptic simulation system for liver diagnosis through palpation [4]. These studies indicate that liver-related training is not limited to cutting. Palpation, localization, and stiffness comparison also have independent training value. Accordingly, the present work does not attempt to reproduce a full hepatectomy workflow, but instead focuses on the more clearly defined and more readily evaluable subtask of hidden-lesion palpation and localization.

Recent robotic palpation systems are more directly comparable to this subtask. They localize hidden lesions through stiffness-adaptive probing or force- and deformation-based sensing [19], [20]. Physical abdominal trainers have also embedded tunable-stiffness nodules and force sensors to support bare-hand palpation practice [21]. In addition, surrogate finite-element models have been used to visualize abdominal tissue stress in near real time for examination training [22]. These systems address the same lesion-search problem but differ from the present work either in modality, by using robot-driven tactile sensing, or in physical form, by relying on a fixed phantom.

C. HAPTIC DEVICE DEGREES OF FREEDOM AND LOW-DIMENSIONAL HAPTIC RENDERING

In medical haptic rendering, device degrees of freedom, force-mapping strategy, and system complexity are tightly coupled. Existing reviews describe both multi-DOF force-feedback interfaces and lightweight low-cost solutions [1]–[3]. For example, Agboh *et al.* proposed a six-DOF laparoscopic training interface, demonstrating the feasibility of high-DOF haptic reconstruction [13]. At the same time, growing evidence suggests that perceptual cues should be selected according to the task goal rather than assuming that full-dimensional, high-fidelity reconstruction is always necessary. Beyond traditional desktop devices, recent work has also proposed a lightweight fingernail-worn haptic device that provides force and vibration cues while minimizing obstruction of the natural fingerpad contact area, offering a new hardware path for low-burden bare-hand interaction [17].

Along this line of reducing haptic dimensionality or emphasizing specific cues, reality-based force profiles have been used to simplify kinesthetic rendering in surgical simulation [14]. These studies suggest that low-DOF haptic output is not merely a simplified substitute, but can also be understood as a task-oriented design choice for highlighting perceptual cues such as normal resistance and variation in stiffness from place to place. Independently of any specific device, benchmarking frameworks have also been proposed to quantify grounded force-feedback performance rather than relying solely on user self-reports [23]. Psychophysical studies of stiffness discrimination under different haptic postures further indicate how reliably users can distinguish the type of stiffness contrast targeted in the present work [24].

D. POSITION OF THIS WORK

In summary, existing studies have established foundations in real-time soft-tissue modeling, liver-related training tasks, and medical haptic rendering. However, there is still room to explore how to effectively integrate a physics engine with a haptic loop under consumer-grade hardware to fulfill the practice-frequency and feedback-quality needs discussed in the Introduction. The GB-cFEM solver used here has previously been validated as a standalone real-time large-deformation algorithm [7], and the fingertip haptic device used here has previously been validated as a standalone lightweight actuator [17]; neither prior study integrated the two with hand tracking into a running, task-oriented closed loop, which is what the present work adds. Based on this gap, this paper develops a task-oriented haptic training prototype using liver lesion localization as the test scenario. The focus is to assess the system integration and identify the engineering boundaries of a physics-computation-driven interaction architecture that aims to provide accessible and physics-rich mechanical cues for medical training. The evaluation in this paper is a formative expert evaluation whose goal is to identify system feasibility and engineering boundaries rather than to demonstrate training efficacy.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. OVERALL SYSTEM ARCHITECTURE

The system consists of four layers: a real-time physics simulation layer, an interaction input layer, a haptic rendering layer, and a task configuration layer. The real-time physics simulation layer uses GB-cFEM to solve the liver tetrahedral mesh in real time. The interaction input layer uses a Stereo IR-170 (Ultraleap, UK) to capture bimanual hand pose, with the five fingertips of the right hand corresponding to five virtual proxy spheres. The haptic rendering layer sends the contact-normal reaction force to a 1-DOF device through a serial protocol. The task configuration layer organizes blind lesion-localization experiments by switching among internal lesion positions.

B. REAL-TIME SIMULATION CORE

GB-cFEM is adopted as the real-time simulation core. The liver tetrahedral mesh is first divided into multiple local groups, each of which is solved independently in a corotational coordinate system for local deformation. During the offline stage, system matrices are calculated. At runtime, the main operations are rotation extraction, right-hand-side construction, and iterative computation. Continuity across groups is maintained through coupling constraints. Compared with a full global implicit FEM, this design avoids reassembling and refactorizing a large linear system every frame. Compared with XPBD, it preserves clearer material-parameter meaning and is therefore better suited to representing stiffness differences between normal tissue and lesion regions.

C. BIMANUAL INPUT AND VIRTUAL PALPATION INTERACTION

The system uses the Stereo IR-170 for bimanual tracking. Fingertips are mapped to capsule-shaped proxy bodies used for local pressing, lesion localization, and surgical-field exposure. The right hand is primarily responsible for palpation and comparison, while the left hand is used mainly to push and retract tissue so as to create a local exposure window for the right hand to enter. To reduce tracking noise and the numerical shock caused by instantaneous penetration, the proxy bodies are not set equal to the raw sensor positions. Instead, they are connected to the device-side positions through damped virtual couplings.

D. 1-DOF HAPTIC RENDERING STRATEGY

The contact-normal reaction force is used as the main input to the 1-DOF device. The fingertip-worn haptic device used here follows the lightweight fingernail haptic design proposed by the authors previously. In this design, a motor drives a fine string to apply normal force on the fingerpad side, while the lightweight actuation mechanism is integrated on the fingernail side to minimize occlusion of the fingerpad and reduce interference with natural touching behavior [17]. Because optical hand tracking has difficulty accurately distinguishing little-finger motion and can generate erroneous feedback, the experimental haptic device was implemented on the four fingers of the right hand, excluding the little finger. On this basis, the device is further integrated into the real-time GB-cFEM simulation loop to convey differences in stiffness during the lesion-search task. The system extracts the contact-force vector $\mathbf{F}_{contact}$ and the surface normal $\mathbf{n}$ from the simulation core and computes the normal reaction-force magnitude delivered to the haptic hardware as

$$F_n = \left|\mathbf{F}_{contact} \cdot \mathbf{n}\right|. \tag{1}$$

Bench characterization with a force sensor measured a maximum static tensile force of 1.36 N. To characterize the frequency response, a DC current bias equal to half the peak-to-peak current was added, and the frequency was swept from 10 to 500 Hz while keeping the peak-to-peak current constant. A resonance was observed near 150 Hz; the peak-to-peak force variation reached 0.434 N at 140 Hz, approximately 3.3 times the 0.132 N measured at 10 Hz. At 500 Hz, the peak-to-peak force variation was 0.123 N, corresponding to 0.93 times the 10-Hz reference value.

The haptic control loop runs at 1000 Hz, with force commands sent over a 2,000,000 bps serial link (under 1 ms per packet). On the default liver mesh (13,800 nodes, 62,897 tetrahedra), profiling the software pipeline over 90 frames gives a mean same-frame latency of 25.2 ms (39.6 FPS, peak 30.1 ms) from hand-input update to force-command issuance, of which 22.2 ms is GB-cFEM/PBD, 1.4 ms is contact and force mapping, and 0.3 ms is rendering. Device forces are additionally low-pass filtered (10 ms time constant) and hand positions are smoothed (30 ms time constant) to suppress transients, and the actuator command path applies a PWM slew limiter to avoid cable-snap artifacts on contact onset. Under this mechanism, users perform lesion localization by perceiving differences in normal resistance while pressing different regions. The hardware connections and wearable device used in the experiment are shown in Fig. 1.

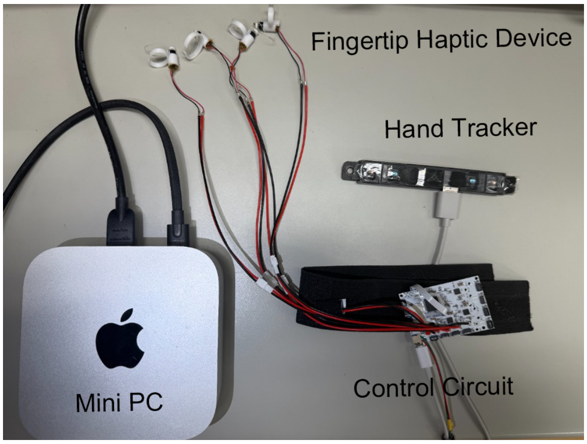

FIGURE 1. Hardware setup used in the experiment. The lower-right part shows the wrist-mounted fixture and driving circuit, while the upper part shows the fingertip haptic actuator. Together with Stereo IR-170 tracking, the system provides hand tracking and normal-force cues.

E. LIVER MODEL AND LESION CONFIGURATION

Under the default configuration used in the physician experience session, the main system parameters are listed in Table 1. These parameters are intended to support a stable and perceptible interactive demonstration and are grounded in the deformation and volume behavior demonstrated by the underlying technical validation. In particular, the ''tetrahedral volume constraint correction'' in Table 1 is not an inherent analytic property of the main GB-cFEM solver. It is an additional stabilization step enabled in the current system. This helps prevent local collapse, abnormal shrinkage, and subsequent contact jitter. The evaluation of the system in this paper therefore comes both from algorithm-level validation data and from the task performance and subjective feedback of a clinical expert during actual interaction.

TABLE 1. Liver Model and Lesion Parameters Under the Default Physician-Experience Configuration

Parameter	Value	Description
Geometry source	NIH 3D (3DPX-021007)	Open liver anatomy model
Meshing parameter	TetGen quality-constrained tetrahedralization (13,800 nodes, 62,897 tetrahedra)	Controls tetrahedral mesh quality and density
Base Young's modulus	7000 Pa	Baseline stiffness of normal liver tissue
Lesion-region Young's modulus	30000 Pa	Local stiffness enhancement in the lesion region
Poisson ratio	0.49	Near-incompressible setting
Number of groups	288	Local grouped-solving configuration
Time step	0.008 s	Real-time simulation step
Tetrahedral volume constraint correction	Enabled (2 iterations, correction = 0.15)	Pulls tetrahedra back toward their rest volume in each step to suppress local collapse, inversion, and abnormal shrinkage under deep pressing

The base and lesion-region Young's-modulus values in Table 1 are chosen to be broadly consistent with reported ranges for normal liver parenchyma and for palpable focal liver lesions. Ex vivo and phantom studies of liver and liver-mimicking tissue report background stiffness on the order of a few to a few tens of kPa [20], [21], and the same body of work reports metastatic or tumor-mimicking inclusions with several-fold higher stiffness than the surrounding tissue [20], [21]. The roughly 4:1 lesion-to-background modulus ratio used here falls within this reported contrast range rather than being set purely for perceptual convenience. The preset lesion size is likewise chosen to be consistent with the clinical size range of focal liver lesions that remain identifiable by manual palpation: a retrospective series of intraoperative liver-metastasis detection found that most nodules subsequently identified by bimanual palpation were on the order of 1 cm or smaller and located at or near the liver surface [25]. To support blind localization experiments, the system presets three switchable local lesion positions inside the liver. Throughout this paper, ''lesion'' refers to this locally stiffness-enhanced tissue region: the term is used in a general sense to denote a mechanically distinguishable abnormal region rather than a specific pathological diagnosis such as a tumor. The lesions are not indicated by visible geometric shape, but are hidden inside the tissue through embedded Young's-modulus enhancement. This forces participants to rely only on local deformation and haptic differences to infer lesion location. In addition to lesion configuration, fixed points are defined in three anatomical regions of the liver model to suppress nonphysical global drift and approximate basic attachment relations: the Porta Hepatis, the Groove for the Inferior Vena Cava (IVC groove), and the Bare Area, as shown in Fig. 3. These regions were chosen because they correspond to where the liver is anatomically tethered by the coronary and triangular ligaments and by the inferior vena cava, which prior work on liver boundary-condition modeling identifies as the dominant anatomical constraints on liver motion [26]. This multi-point, surface-based anchoring strategy approximates the stable connection between the liver and major vascular and diaphragmatic attachment regions while preserving local deformability. Consistent with that same prior work, it is a simplified fixed-boundary approximation, which is a source of boundary-condition error relative to a fully patient-specific simulation.

Beyond these anatomical anchors, a static liver-shaped abdominal support boundary is also introduced to approximate the external constraints imposed by surrounding abdominal tissue. Specifically, the system extracts the contact surface from the outer triangular mesh of the liver in the rest pose and offsets it outward along the surface normal of each facet by a distance proportional to the bounding-box diagonal, thereby constructing an envelope shell that follows the liver geometry while being slightly expanded. This shell is not designed as a closed abdominal box. A local opening is retained along a selected direction and serves as an exposure window created after incision and retraction, allowing fingers to enter from the exposed side for palpation.

At runtime, contact between surface vertices and this static shell is computed through local adjacent-triangle search and closest-point projection. Each vertex is required to stay on the interior side of the shell. When the projection indicates that a vertex has moved to the exterior of the shell by more than a small threshold along the local outward normal, the system applies a push-back correction along that normal, combined with tangential damping and velocity update to suppress oscillation. Accordingly, the ''abdominal collision'' in this paper should be understood more precisely as a liver-shaped support envelope generated from the rest-pose geometry rather than a high-fidelity anatomical reconstruction of the abdominal wall, diaphragm, and surrounding organs. This is also one of the main limitations of the current model. Although it provides more plausible spatial support than a simple planar wall, it does not yet model the anterior falciform ligament or authentic abdominal contact relations individually, nor does it represent ligament compliance the way a patient-specific boundary-condition model would [26]. The runtime interface and the division of left-hand exposure and right-hand palpation are shown in Fig. 2.

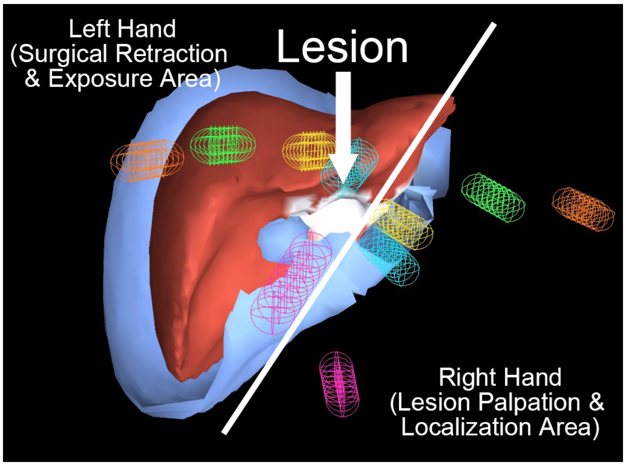

FIGURE 2. Runtime screenshot of the system. The red region is the liver model, and the outer light-blue shell is the liver-shaped collision cavity used to provide spatial constraint. One side is retained as an open region to approximate the local exposure window in open surgery. The colored wireframe capsules denote fingertip proxy bodies. The left-side region corresponds to the left hand, used for pushing and retracting tissue to maintain exposure, while the right-side region corresponds to the right hand, used for lesion palpation, pressing-based comparison, and localization search.

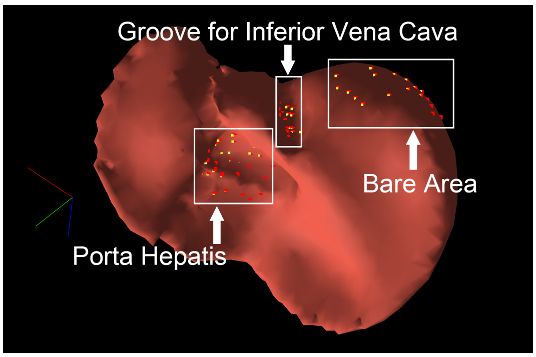

FIGURE 3. Anatomical fixation regions and fixed-node distribution in the liver model. The dotted markers denote node clusters with fixed-point constraints, corresponding to the Porta Hepatis, the Groove for the Inferior Vena Cava, and the Bare Area. These constraints approximate major anatomical attachments and suppress nonphysical rigid-body drift.

IV. EXPERIMENTAL DESIGN AND EVALUATION METRICS

A. APPLICATION-LEVEL TECHNICAL VALIDATION BASED ON A SPECIFIC LIVER MODEL

Although the systematic benchmark of the underlying pure algorithmic GB-cFEM framework, such as idealized tests on basic geometries, has already been published in [7], applying it to an actual medical model within a haptic loop imposes new requirements on real-time performance and physical stability because of increased model complexity. Therefore, instead of relying on the generic 3D models used in [7], this paper re-runs validation on the liver anatomical model used by the present system in order to show that the computational framework can still serve as the real-time basis of the current interactive system:

1) Large-deformation accuracy: displacement response of the proposed method and XPBD is compared under unified material parameters and loading conditions, using VegaFEM as the high-accuracy reference.
2) Near-incompressible volume preservation: under large dragging with different Poisson-ratio settings, the maximum and mean deviations of the volume ratio V/V_0 are measured.
3) Anisotropic response validation: force-displacement slopes under loading in different directions are compared to examine whether the system can stably convey direction-dependent mechanical differences.
4) Real-time performance and scalability: the physics-solver frame rate is measured at different tetrahedral scales to assess interactivity under standard CPU conditions.

In this paper, "XPBD Fast" refers to a typical real-time configuration with low substeps (subSteps $= 5$, maxIterations $= 1$, maxIterationsV $= 5$), whereas "XPBD Ref" refers to a higher-accuracy reference configuration under the same mesh and the same fixation strategy but with the number of XPBD substeps increased tenfold to 50 (maxIterations kept at 1) in order to observe its error and cost near convergence.

Both XPBD configurations are reported to ensure a fair evaluation. Reporting only a low-substep setup would unfairly penalize XPBD's accuracy, while reporting only a high-substep setup would obscure its computational cost. Including both clarifies the inherent tradeoff of XPBD between real-time performance and deformation accuracy, thereby highlighting the relative advantages of the proposed method.

Table 2 lists the solver-level settings shared or contrasted across the three methods in the displacement-accuracy and volume-preservation comparisons. All three methods were run on the same TetGen-generated liver mesh and the same deterministic fixed-vertex definition exported from the interactive system.

TABLE 2. Solver Settings Used in the Displacement-Accuracy and Volume-Preservation Comparisons

Setting	Proposed method	XPBD Fast / XPBD Ref	VegaFEM
Time step	0.01 s	0.01 s	0.01 s
Solver iterations per step	Coupling iterations: 30 / 150	Substeps: 5 / 50 (maxIterations = 1 per substep)	Newton-type iterations: 1 / 2 (implicit backward Euler)
Baseline material	$E = 7000$ Pa, $\nu = 0.49$	$E = 7000$ Pa, $\nu = 0.49$	$E = 7000$ Pa, $\nu = 0.49$
Deformation model	Grouped corotational FEM, 288 groups	XPBD FEM-based solid model	Corotational linear FEM (CLFEM)

For the large-deformation accuracy comparison (Experiment 1), a ramped pull force was applied to a deterministic frontal target region. The influence radius was set to 0.6 of the local bounding-box scale. Three loading strengths were tested, with relative magnitudes of 16, 30, and 40. For the proposed method and XPBD, each trial followed a settle (120 steps) $\rightarrow$ load-ramp (240 steps) $\rightarrow$ hold (240 steps) protocol. VegaFEM's implicit solver is substantially more expensive per step. Therefore, its schedule was proportionally shortened to settle (60 steps) $\rightarrow$ load-ramp (120 steps) $\rightarrow$ hold (120 steps). The same three loading strengths were retained. The terminal, post-hold displacement was used for comparison. This ensured that the reported error reflected converged behavior rather than an unconverged transient.

For the near-incompressible volume-preservation comparison (Experiment 2), an anchor slice covering the posterior 5% of the model was fixed while an anterior pull slice (covering the anterior 12% of the model, at least 24 vertices) was dragged by an imposed displacement, following the same settle (120) $\rightarrow$ drag (240) $\rightarrow$ hold (240) schedule for all three methods. The Poisson ratio was swept between a baseline value ($\nu = 0.28$) and a near-incompressible value ($\nu \approx 0.47$–0.49) while $E = 7000$ Pa was held fixed, and the resulting volume ratio V/V_0 was recorded on the same mesh for all three methods.

For both Experiment 1 and Experiment 2, the termination criterion for each run is a fixed step count: metrics are read only after the hold phase, at which point the displacement and volume ratio of the proposed method and of XPBD have already visibly plateaued. Because VegaFEM's implicit Newton solver converges within each individual step, its per-step iteration count in Table 2 instead reflects an inner Newton-convergence budget, whereas the proposed method and XPBD use a fixed number of outer coupling iterations or substeps per step without an explicit residual threshold. The reported comparisons use steady-state values after transients under a fixed-step schedule, not a common convergence tolerance across methods.

Unless otherwise stated, all system runtime tests, real-time demonstrations, and the formative expert evaluation in this paper were conducted on the same Mac mini (Apple, USA) with an Apple M4 Pro (12-core CPU), 24 GB memory, and macOS 26.3.1. This is emphasized because the goal of the paper is not only theoretical real-time performance, but also actual deployability on compact consumer-grade hardware. As a supplementary whole-machine observation, when the system ran continuously for 5 minutes on the same device, the wall power averaged about 52 W, compared with about 12 W at idle. This low power consumption provides an additional dimension for hardware feasibility assessment, serving as strong evidence of the low computational burden of the real-time FEM interaction pipeline, which requires only several tens of watts rather than a high-power workstation load.

B. FORMATIVE EXPERT EVALUATION AND OBSERVATION OF LESION-SEARCH BEHAVIOR

One surgeon participated in the system evaluation. At this stage of development, a single-expert, walkthrough-based formative evaluation was chosen. Before investing in a structured multi-participant protocol, it was necessary to first determine whether the simplified anatomical constraints, the current material model, and the 1-DOF force channel could even produce mechanically and clinically sensible feedback. Running a large novice-and-expert study prior to this engineering diagnosis would have been premature and resource-inefficient. This staged strategy follows guidance that a simulation should first be piloted and iteratively refined before a larger validity study is undertaken [27], [28], and the view that validity is not a fixed property of a simulator but must be argued for a specific intended use [29], [30]. Prior negative results with abdominal-palpation simulators also caution that even a well-designed simulated task does not automatically yield a reliable competency measure [31]. The participant profile is listed in Table 3. The study protocol was approved by the ethics committee of the Institute of Science Tokyo (approval number: 2023362). The participating surgeon was informed about the study before the experiment and signed an informed-consent form.

This evaluation was a formative expert evaluation. Its goal was to identify system feasibility and engineering boundaries, not to measure training effectiveness, skill acquisition, or learning outcomes; therefore, any conclusions regarding these outcomes should be considered preliminary. The evaluation setup is shown in Fig. 4. The expert first watched an approximately 5-minute system demonstration and then completed, in sequence, free exploration, a blind lesion-localization observation, and a usability evaluation task.

In the lesion-localization task, to reduce spatial-memory effects, different unknown lesion positions were used under the two conditions of haptics disabled and haptics enabled. The system presets three switchable lesion locations inside the liver (Section III-E); all three share the same local radius and the same Young's-modulus enhancement, so the two trial positions were matched in lesion size and stiffness contrast by construction. However, because the three presets were placed in different anatomical regions, differences in depth and local geometry may have affected task difficulty. The presets are also cycled in a fixed, deterministic order, so a learning or familiarization effect between the haptics-off and haptics-on trials cannot be excluded either. The strategy change reported below should therefore be viewed as a plausible and suggestive observation, while the specific contribution of haptic feedback remains to be confirmed.

Given the single-session, diagnostic scope of this formative round, the task was not instrumented to automatically record quantitative indicators such as localization accuracy, time-to-localization, number of exploratory presses, hand trajectory, or error distance. Beyond the lack of instrumentation, several indicators are difficult to define for continuous, bimanual exploratory palpation. For example, a "press" has no clear start or end when the expert alternates between sliding, sustained compression, and brief taps. Similarly, "error distance" assumes a specific target point. In our simulation, however, the lesion is a diffuse region of increased stiffness rather than a point target. Because the two trials differed in lesion position and order, these single-trial values would provide only a confounded $n = 1$ comparison. The expert's search behavior and strategy changes under the two conditions were instead recorded qualitatively, through experimenter observation and think-aloud verbal feedback, for analysis of how the local stiffness cues provided by the system supported palpation judgment. We consider this a genuine limitation of the present evaluation. The interaction pipeline already records per-frame fingertip proxy positions and contact states for haptic-force computation. The next step is to define these indicators operationally and enable their logging. This should be evaluated in a structured, randomized, multi-participant study with a controlled haptics-on/off comparison.

TABLE 3. Profile of the Surgical Expert Participating in the Formative Evaluation

Attribute	Information
Professional title	Intermediate
Clinical experience	7 years
Specialty	Surgery
Surgery-related experience	2 years

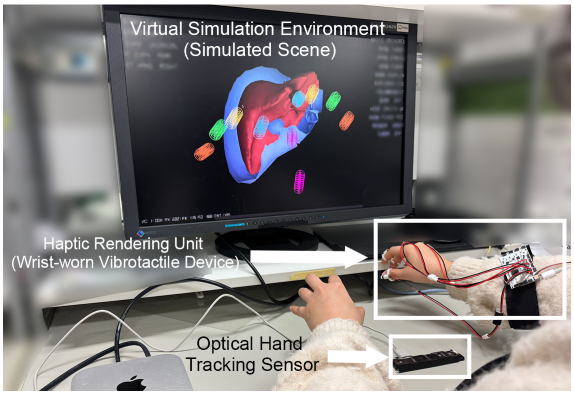

FIGURE 4. Formative expert evaluation setup. The participant performed virtual palpation through hand motion while receiving 1-DOF force feedback, and the screen displayed real-time liver deformation and proxy-body positions.

The task design and recorded contents of the formative expert evaluation are listed in Table 4.

TABLE 4. Task Design and Recorded Contents of the Formative Expert Evaluation

Task	Goal	Condition	Recorded content
Task A: Free exploration	Observe interaction fidelity and naturalness	Freely touch the liver while verbalizing thoughts	Subjective evaluation and interaction behavior
Task B: Blind lesion localization	Observe search-behavior changes across two trials	Trial 1: haptics off; Trial 2: haptics on with a different lesion position	Operational strategy and verbal feedback
Task C: System usability feedback	Evaluate engineering boundaries and physical consistency	Press different regions and compare resistance differences	Semi-structured interview and Likert-scale ratings

C. METRIC CONSTRUCTION FOR THE FORMATIVE EVALUATION

The formative evaluation in this paper records three kinds of indicators: behavioral indicators, subjective evaluation, and rating-scale scores. The assessment emphasizes qualitative observation and system-mechanism analysis in order to provide diagnostic and design guidance for the next generation of the system.

V. RESULTS

A. TECHNICAL VALIDATION RESULTS

1) Large-Deformation Accuracy and Real-Time Performance

When the test was re-run on the real liver model used in this paper, with VegaFEM taken as the reference, Fig. 5 and Table 5 show that the kernel used by the present system achieved an average displacement error of about 17.9% under three loading strengths, whereas the error of XPBD Fast exceeded 50%. More importantly, for the liver-lesion palpation scenario considered here, which contains tens of thousands of tetrahedra, the method still achieved about 83 FPS at a computational cost of approximately 12 ms per frame, satisfying the real-time interaction requirement of the system. Although XPBD reached higher FPS at low substeps, its displacement response was overly soft, whereas the higher-accuracy configuration sacrificed real-time capability. Considering that the system test was carried out on a Mac mini M4 Pro and that the average wall power of the full machine was approximately 52 W during continuous 5-minute operation (versus about 12 W at idle), this low power consumption provides a metric for hardware feasibility. It indicates that the FEM-driven pipeline maintains a low computational burden, thus it is not limited to heavy computing platforms and can support actual interactive operation with only several tens of watts of additional whole-machine power.

TABLE 5. Comparison of Large-Deformation Accuracy and Real-Time Performance

Metric	Proposed method	XPBD Fast	XPBD Ref	VegaFEM
Mean displacement error	17.9%	50.8%	43.6%	0% (reference)
Mean runtime	12.0 ms	7.9 ms	59.2 ms	–
Frame rate	83.3 FPS	126.6 FPS	16.9 FPS	–

Taken together, this result provides supporting evidence for the following judgment: the simulation kernel relied upon by the present system does not simply trade accuracy for speed, but maintains a more stable stiffness representation than XPBD within an interactive operating range.

2) Near-Incompressible Volume Preservation

Specialized validation under the typical near-incompressible setting of a high-water-content soft tissue such as the liver shows that, as presented in Fig. 6 and Table 6, the volume stability of the kernel used here is overall better than that of XPBD and close to the VegaFEM reference. In particular,

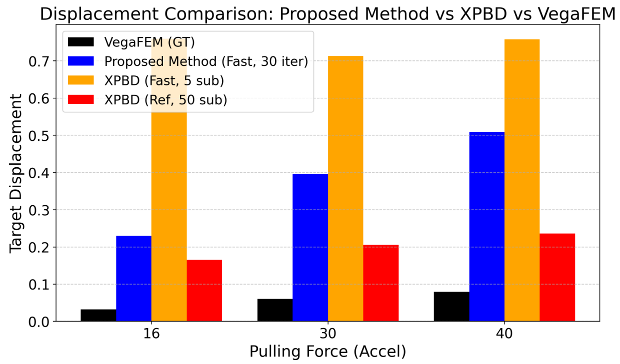

FIGURE 5. **Comparison of large-deformation displacement. Relative to the VegaFEM reference, the proposed method exhibits more stable displacement response than both XPBD Fast and XPBD Ref under all three loading strengths.**

under the $\nu = 0.47$ condition, the mean volume deviation of the tested liver model was 0.92%, and the maximum deviation was 1.35%, both better than XPBD.

TABLE 6. **Volume-Deviation Comparison Under Near-Incompressible Conditions**

Method	Poisson ratio	Maximum volume deviation	Mean volume deviation
Proposed method	0.28	1.88%	1.31%
Proposed method	0.47	1.35%	0.92%
XPBD	0.28	7.27%	1.84%
XPBD	0.47	1.59%	1.29%
VegaFEM	0.28	1.94%	1.28%
VegaFEM	0.47	1.47%	0.80%

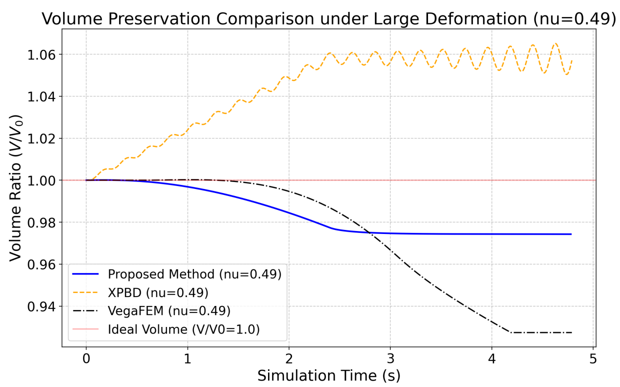

FIGURE 6. **Comparison of volume preservation under near-incompressible conditions. The proposed method maintains small volume deviation under both Poisson-ratio settings and is overall close to the VegaFEM reference.**

This indicates that, under the current protocol, the proposed method is less likely to produce the appearance of abnormal shrinkage or collapse under large pressing and dragging, thus providing a more consistent geometric basis for subsequent haptic mapping.

3) Real-Time Performance and Scalability

Following the performance framework of [7], we further evaluated the method at different mesh scales. For all three methods, tetrahedral meshes were generated from the same liver surface model. The target sizes ranged from approximately 1,000 to 65,000 tetrahedra. We varied the TetGen mesh-quality and maximum-tetrahedron-volume parameters, and tuned the volume bound to reach each target size. Each configuration was warmed up before measuring frame time. For the proposed method, we used 60 warmup frames and 240 measurement frames. Comparable schedules were used for XPBD and VegaFEM. The proposed method's coupling-iteration count was held fixed at 10 per frame across all scales, and both single-thread and 10-thread configurations were measured on the same 12-core CPU to characterize multi-core scalability. The XPBD measurement reported here used the higher-accuracy "XPBD Ref" configuration defined above (50 substeps, maxIterations = 1) rather than the low-substep "XPBD Fast" configuration, which is why its frame rate is substantially lower than a real-time-tuned XPBD setup would achieve at the same mesh scale. As shown in Fig. 7 and Table 7, the simulation kernel reached 219.24 FPS at approximately 7,790 tetrahedra and still maintained about 20.2 FPS at approximately 35,486 tetrahedra. By contrast, VegaFEM achieved only 5.52 FPS at about 18,245 tetrahedra, and the high-accuracy XPBD reference configuration achieved 16.27 FPS at about 20,061 tetrahedra.

TABLE 7. **Comparison of Real-Time Performance Across Mesh Scales**

Method	Mesh scale	FPS	Note
Proposed method	7,790 tets	219.24	Above real-time interaction demand
Proposed method	35,486 tets	20.2	Maintains basic interactivity
XPBD Ref	20,061 tets	16.27	High accuracy comes at high cost
VegaFEM	18,245 tets	5.52	Difficult to use interactively

Although the mesh scales used by different methods are not matched point by point, this result is still sufficient to indicate that the GB-cFEM route used by the current system can support medical demonstration and interactive tasks on a consumer-grade CPU without relying on heavier high-end computing conditions.

B. RESULTS OF THE FORMATIVE EXPERT EVALUATION AND SYSTEM-USABILITY FEEDBACK

1) Interaction Fidelity During Free Exploration

During the free-touch task, the expert participant first pressed different parts of the liver surface with multiple fingers, then attempted to lift the thin edge of the liver, and further explored the dynamic change of force feedback by varying the pressing intensity. Based on comparative palpation between the preset lesion region and normal tissue, the expert identified the

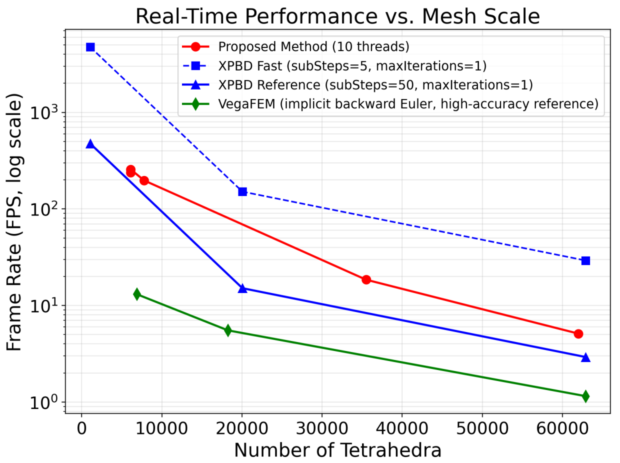

FIGURE 7. Frame rate versus mesh scale (log-scale FPS axis) for the proposed method (10 threads), XPBD under the Fast and Reference substep configurations defined in Section IV-A, and VegaFEM. The proposed method maintains better real-time performance than VegaFEM over a broad tetrahedral range and still retains basic interactivity on larger-scale models.

engineering limitations of the system from both anatomical and physical perspectives.

The first issue concerned distortion of material behavior. The expert pointed out that the current liver soft-tissue model was overly elastic, with a feel closer to homogeneous soft gel than to fibrous biological tissue. A real liver exhibits nonlinear stiffness and is covered by a capsule, producing an initial resistance followed by constrained compression at deeper indentation. By contrast, the present model deforms too easily under compression.

The second issue concerned local perceptual differences caused by simplified anatomical constraints. The expert observed that when the liver was pushed laterally, the model exhibited global translation similar to a floating object rather than rotation constrained by anterior ligaments. In the current physical model, boundary conditions were simplified using an equivalent surface-based fixation strategy. Fixed points were assigned directly to the Porta Hepatis, the Bare Area, and the Groove for the Inferior Vena Cava to provide basal locking, superior adhesion, and posterior vertical support, respectively. Although this macroscopic fixation ensures physical stability under large deformation, the expert's feedback indicates that the lack of anterior traction constraints still creates a gap between simulated feedback and clinical expectations during pushing maneuvers.

The final issue concerned deviation from the clinical interaction paradigm. The expert noted that bare-hand palpation is part of clinical diagnosis, but in surgical contexts the main interaction medium is a surgical instrument. Direct hand-tracking interaction sacrifices part of the task validity of surgical simulation and makes the experience closer to physical examination than to operative practice.

2) Behavioral Observation in the Lesion-Localization Task

In the initial exploration with haptic feedback disabled, the expert relied mainly on visual deformation for judgment. Observation showed that the expert first attempted light surface pressing and, after failing to obtain useful stiffness cues, gradually shifted to larger pulling motions and deeper indentation.

In the localization test with haptic feedback enabled, where the lesion was placed in a new unknown region, the expert's search behavior changed to local comparative light pressing. Verbal feedback indicated that the local differences in normal resistance provided by haptic feedback supported discrimination of the lesion. This strategy change suggests that the FEM-driven stiffness-mapping logic can provide perceptible physical cues for the lesion-search task. Because the two trials used different lesion positions in a fixed order, and because no quantitative localization metrics were recorded (Section IV-B), this observed change cannot be isolated from possible lesion-difficulty or learning-order effects. It is therefore reported as a suggestive qualitative trend rather than a controlled comparison. The expert's operational feedback reveals the boundary between this engineering prototype and a clinical trainer. These qualitative observations play a diagnostic engineering role by providing concrete experiential guidance for future improvements in medical-simulation fidelity.

3) Diagnosis of Engineering Constraints and Evaluation of Force-Rendering Quality

In Task C, the expert actively performed continuous sliding exploration and deep pressing on the liver surface while cross-checking fingertip force sensation against the visual deformation shown on the screen. Based on these dynamic palpation behaviors, the expert identified a core engineering bottleneck in the system connectivity: the hardware limitation in feedback directionality. The current 1-DOF output can convey only one-directional normal resistance, while lacking tangential force and more complex reaction-force distribution. As a result, it cannot reproduce the enveloping squeeze sensation that a real finger experiences when pressed into tissue and surrounded by the material. This reflects an inherent limitation of low-dimensional hardware. The corresponding Likert-scale ratings are listed in Table 8.

TABLE 8. Likert-Scale Ratings of System Usability and Force-Rendering Quality

Evaluation dimension	Score (out of 5)	Interpretation
System feasibility and potential	4/5	The "continuum mechanics + haptics" system architecture was considered feasible
Soft-tissue mechanical expression	4/5	The stiffness settings of the physical engine could be effectively mapped into haptic cues
Force-feedback realism	3/5	The physical limitation of single-DOF hardware dimensionality and refresh behavior was revealed

According to the Likert-scale scores in Table 8, the system received positive ratings in ''system feasibility and potential'' and ''soft-tissue mechanical expression'' (4/5), while the score for ''force-feedback realism'' was only moderate (3/5). This result confirms the successful integration of the system's data flow while identifying future engineering challenges in hardware frequency response and feedback dimensionality. Because these scores come from a single surgical expert, they are reported only as qualitative, formative feedback that corroborates the verbal walkthrough findings; consistent with guidance on interpreting ordinal rating data [32], they are not statistically generalizable usability evidence and should not be interpreted as validating training effectiveness.

VI. DISCUSSION

A. FROM ALGORITHM VALIDATION TO SYSTEM-PROTOTYPE EVALUATION

As established in the Position of This Work discussion, the present contribution is the task-oriented integration of the previously separate GB-cFEM solver and fingertip haptic device into a medical palpation scenario, evaluated here through application-level technical validation and formative expert diagnosis. The technical validation provides supporting evidence that the simulation kernel can support real-time stiffness expression on a medical model, while the formative evaluation identifies the boundaries of the model as an interactive prototype through walkthrough-based diagnosis. This single-expert walkthrough was formative and diagnostic. It focused on identifying engineering limitations and assessing whether the system's physics-driven cues were mechanically and clinically plausible. The observations below therefore provide preliminary qualitative evidence rather than a formal evaluation of training efficacy, educational impact, or learning outcomes. The discussion of medical-education needs in the Introduction is intended to motivate the potential value of the system. Its educational effectiveness will be examined in the dedicated novice-and-expert study outlined in the Future Research Directions subsection.

The three issues raised by the expert in Section V-B, namely overly elastic material behavior, floating-object-like global translation under lateral pushing, and the directional limitation of 1-DOF output, are not merely generic engineering limitations. They directly affect the judgment logic of the core lesion-localization task. The absence of material nonlinearity weakens the operator's ability to probe deep lesions based on progressive resistance. The simplified anatomical constraints let organ-level displacement mask local hardness abnormalities near the margins. The lack of tangential cues in the 1-DOF channel forces the operator to rely on dense vertical pressing rather than sliding to outline lesion boundaries. Together, these findings show how engineering simplifications can alter palpation behavior in ways not captured by numerical tests alone. They also define concrete targets for the next iteration of the system.

To avoid conflating these issues, it is useful to separate them into two distinct dimensions of realism. The current prototype does not claim to have fully resolved either dimension. The first dimension is anatomical realism. It describes how accurately the model represents the liver and its surrounding structures. In the current system, the liver is modeled as a homogeneous, nearly incompressible, linear-elastic body without a separate capsule layer. Its ligament attachments are simplified into three fixed regions, and the surrounding abdominal environment is represented by a resting-state envelope shell. The second dimension is haptic realism. It describes how accurately the system reproduces the forces felt during real palpation. The 1-DOF actuator provides only normal contact force at each fingertip. It cannot reproduce tangential force during sliding, the squeezing sensation around the fingertip, or distributed feedback such as friction, texture, and differences among multiple contact points. The absence of material nonlinearity and capsule behavior affects both dimensions. Anatomically, it limits the fidelity of the tissue model. Haptically, it weakens the progressive increase in stiffness that a real liver would produce during deep pressing. Both dimensions were deliberately simplified to preserve real-time performance and compatibility with consumer-grade hardware. They remain open engineering targets rather than established capabilities of the current system.

B. APPLICATION FOCUS AND FUTURE EVOLUTION OF THE SYSTEM

For the test scenario of liver lesion localization, the current system expresses lesion characteristics through normal-force-based stiffness feedback. At the same time, what is validated here is not an isolated implementation for one specific liver model. Instead, it is a transferable way of organizing the system. The "GB-cFEM kernel + region-varying material stiffness + haptic mapping" architecture may extend to other soft-tissue palpation or lesion-search scenarios that rely on regional stiffness differences, contact-reaction cues, and real-time soft-tissue deformation. However, such extensions require organ-specific changes to the geometry, boundary constraints, material parameters, and interaction paradigm.

C. FUTURE RESEARCH DIRECTIONS

Future work should involve a structured, multi-participant user study with both medical novices and experienced surgeons to examine task validity and training effectiveness, focusing on quantitative learning curves and skill transfer, although the current scope remains focused on system-prototype design and technical feasibility. On the engineering side, the next iteration will prioritize a nonlinear, capsule-aware material model, more complete ligament and abdominal boundary constraints for stronger organ anchoring, integration of surgical-instrument proxies to correct the interaction paradigm, and improved tangential-force compensation and synchronization between the physics engine and the haptic loop.

VII. CONCLUSION

This paper proposes and implements a task-oriented haptic training prototype to verify the system connectivity and engineering feasibility of integrating a GB-cFEM physical engine with a haptic interface. By combining GB-cFEM-based real-time soft-tissue simulation with haptic mapping and using liver lesion localization as a concrete test scenario, the study realizes the transformation of a physical simulation kernel into an interactive feedback engine.

The technical validation provides application-level evidence regarding robustness, real-time performance, and low computational burden on a medical model, showing that the system can run stably on ordinary consumer-grade hardware with low power consumption. However, these comparative results should still be understood as comparative validation for a system prototype rather than as a standardized cross-method benchmark. The evaluation in this paper is a formative expert evaluation whose goal is to identify system feasibility and engineering boundaries rather than to prove training effectiveness. The assessment by a single surgical expert identifies the engineering challenges and boundaries that must be addressed before the prototype can move toward a high-fidelity clinical trainer, including richer nonlinear material properties, more complete anatomical constraints, and expanded feedback dimensionality.

In summary, this study demonstrates that a physics-driven, GB-cFEM-based haptic simulation loop can be engineered to run in real time on compact, consumer-grade hardware, and that a surgical expert could extract physically meaningful stiffness cues from it during a brief walkthrough. These results support the technical feasibility and system-level connectivity of the proposed architecture, and they suggest, without by themselves demonstrating, that such systems could eventually help address the hardware-accessibility and high-information-feedback needs of high-frequency medical training. Establishing training effectiveness, learning-efficiency gains, and clinical-grade fidelity will require the structured, multi-participant study involving both novices and experts outlined in the Future Research Directions subsection. The present single-expert formative evaluation nevertheless provides preliminary evidence and useful insights into these aspects, although further validation is needed.

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